

Whole Exome Sequencing Analysis

Patient name: XX	PIN: XX
Age: 32 years	Gender: Male
Hospital/Clinic: XX	Sample Number: XX
Referring Clinician: XX	Sample Collection Date: NA
Specimen: Peripheral Blood	Sample Received Date: NA
	Report Date: NA

Clinical History

Proband XX presented with a history of progressive visual impairment, characterized by difficulty seeing in low-illumination settings (nyctalopia/night blindness) and gradual blurring of vision during daytime. The clinical features are consistent with a diagnosis of retinitis pigmentosa. Proband Mr. XX, has been evaluated for pathogenic variations.

Results

Likely compound heterozygous variants causative of the reported phenotype was identified in *EYS* gene

Summary of Findings	
Potentially Relevant Variant(s):	Likely Pathogenic variant (c.5825delG) and Variant of Uncertain Significance (c.1185-6T>G) were identified in <i>EYS</i> gene
Additional variant(s):	No Pathogenic or Likely Pathogenic variants were detected.
CNV Findings	
No significant Copy Number Variations (CNV) related to phenotype was detected.	

List of significant variants identified related to the phenotype:

Variant :1

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
EYS (-)	c.5825delG (p.Gly1942ValfsTer3)	Autosomal Recessive	Retinitis pigmentosa 25 (OMIM# 602772)	PM2,PVS1	Likely Pathogenic
Region	Allele Status	Sequence Ontology	Variant Frequency	Allelic Read Depths	
Exon 27	Heterozygous	Frameshift	Not observed in annotated gnomAD v4 All	Ref(C): 34, Alt(-): 45, VAF: 56.96%	
In silico Prediction Tools				Conservation Status	
LRT: -		CADD: U	MT2: D	PhyloP: C	GERP++: -
Genomic Position			Clinical & Literature Evidences		
Chr6:NC_000006.12:g.64439172delC (NM_001142800.2)			-		

Variant :2

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
EYS (-)	c.1185-6T>G (Intronic variant)	Autosomal Recessive	Retinitis pigmentosa 25 (OMIM# 602772)	PM2,PM3	Uncertain Significance
Region	Allele Status	Sequence Ontology	Variant Frequency	Allelic Read Depths	
Intron 7	Heterozygous	Intron	0.0035% frequency observed in Annotated gnomAD v4 South Asian background	Ref(A): 46, Alt(C): 48, VAF: 51.06%	
In silico Prediction Tools				Conservation Status	
GeneSplicer: D		MaxEntScan: D	CADD: U	PhyloP: C	GERP++: C
Genomic Position			#Clinical & Literature Evidences		
Chr6:NC_000006.12:g.65384506A>C (NM_001142800.2)			ClinVar: 963338 PubMed: 29159838		

Abbreviation: **D:** Deleterious; **Da:** Damaging; **Ds:** Disrupted; **C:** Conserved; **T:** Tolerated; **U:** Uncertain

#Clinical & Literature Evidences: This variant has been classified as Likely pathogenic in ClinVar database and has been previously reported in a patient affected with retinitis pigmentosa in compound heterozygous state.

OMIM Phenotype: Retinitis pigmentosa 25 (OMIM#[602772](#)) is caused by homozygous or compound heterozygous mutation in the EYS gene (OMIM#[612424](#)). This disease follows autosomal recessive pattern of inheritance [2].

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].

No other variant that warrants to be reported for the given clinical indication was identified

The variants are reported to be in likely compound heterozygous state. These likely compound heterozygous variants are strongly recommended to confirm the Cis or Trans status by parental segregation analysis.

CNV Findings

No significant Copy Number Variations (CNV) related to phenotype was detected.

Recommendations

- Sequencing the variant(s) in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Alternative test is strongly recommended to rule out the deletion/duplication.
- Genetic counselling is advised.

Methodology

DNA extracted from the blood was used to perform whole exome using whole exome capture kit. The targeted libraries were sequenced to a targeted depth of 80X to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. In-silico predictions of the variant was carried out using VS- SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported. Gene list used in the assay [Click here](#).

Sequence data attributes

Total Read Generated	9.37 Gb
Data ≥ Q30	97.54%

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR and Promoter region variants are not assessed using this assay.
- This assay has a sensitivity of 70-75% in detecting large deletions/duplications of more than 10 base pairs or copy number variations (CNV). However, it is important to note that any CNVs detected must be confirmed using an alternate method.

- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions. The variants of uncertain significance and variations with high minor allele frequencies which are likely to be benign will be given upon request.
- In addition, this test does not cover detection of such structural variants in the genes *SMN1*, *SMN2*, *HBA1*, *HBA2*, *PMS2*, *GBA*, and *CYP21A2* which have very high homology. Moreover, the ORF15 transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].

References

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genetics in medicine: official journal of the American College of Medical Genetics. 17.5 (2015): 405-424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). Nucleic Acids Res. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. Lee, Kristy et al. "ACMG SF v3.3 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." Genetics in medicine: official journal of the American College of Medical Genetics vol. 27,8 (2025): 101454. doi:10.1016/j.gim.2025.101454.

This report has been reviewed and approved by:

			
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